EET202

HINES & TRANSFORMERS

MODULE:1

re:1 – Constructional Details

edge about constructional details of DC machines

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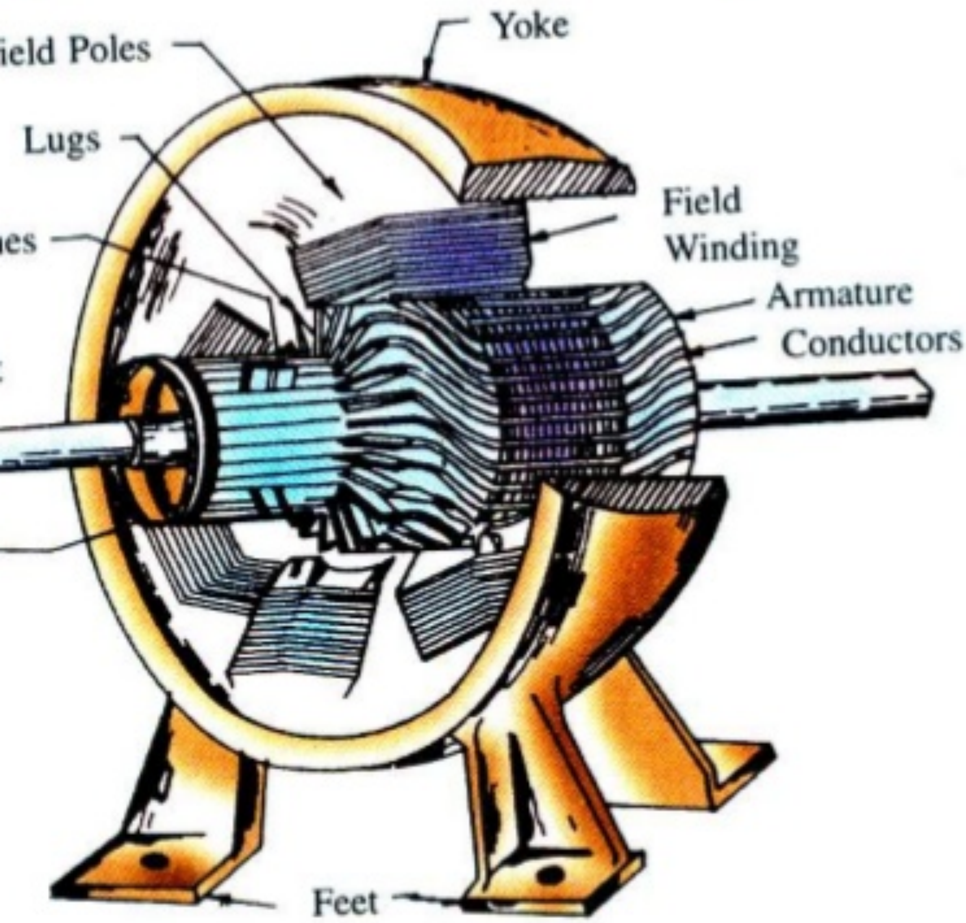
Module – 1

of dc machines - armature winding- single layer winding, double
wave, equalizer rings, dummy coils, MMF of a winding, EMF
etic torque - numerical problems.

nstructional Details

DC Machine



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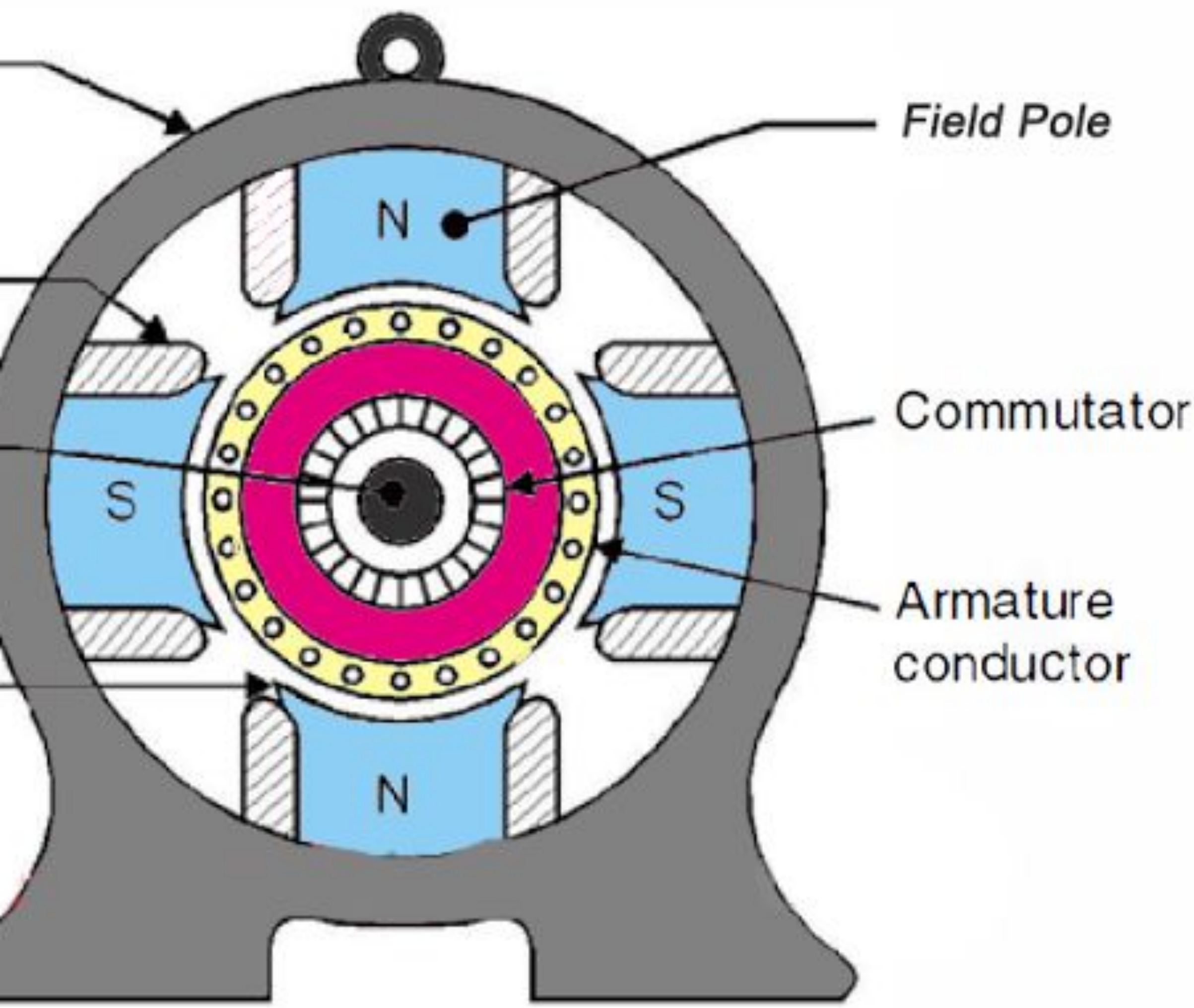

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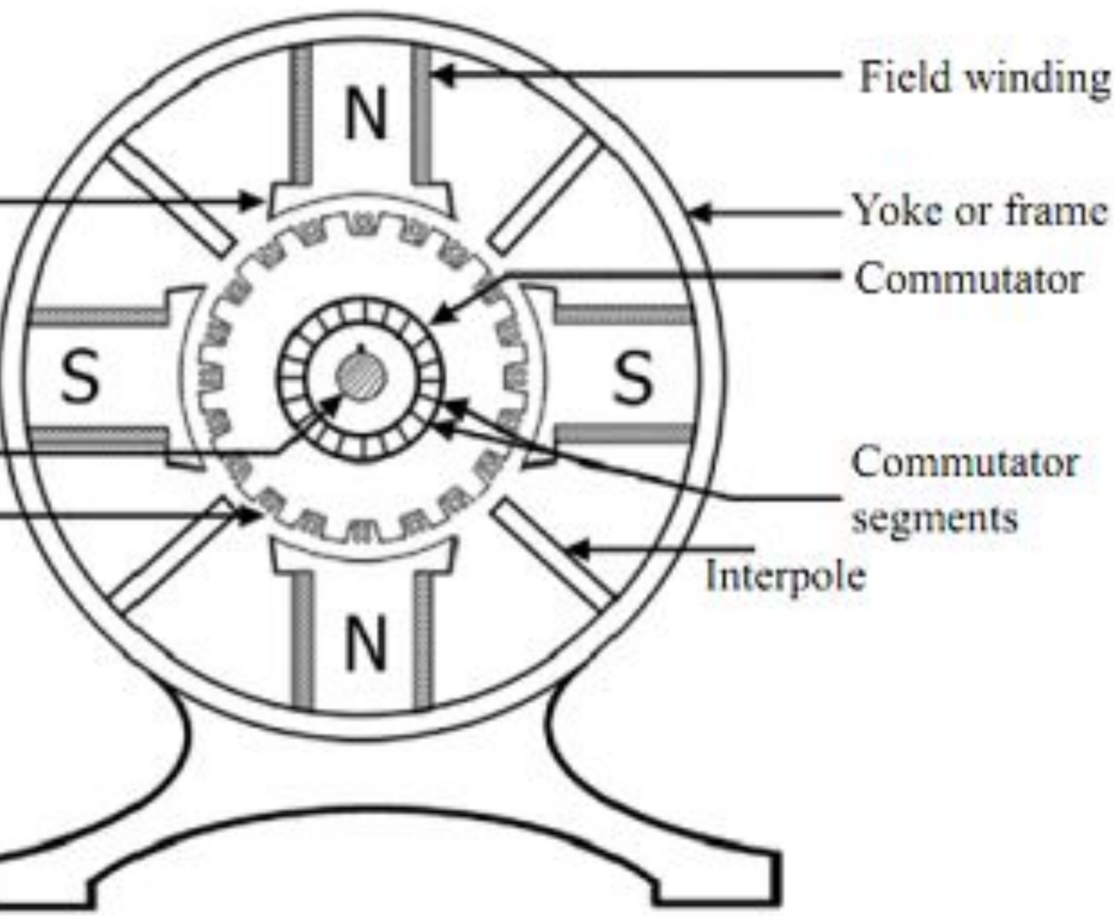


nstructional Details

ts are:

ke or Frame

eld System

- Pole Core
- Pole Shoe
- Field Coils / Field Windings

mature

- Armature Core
- Armature Windings

mmutator

ushes

nstructional Details

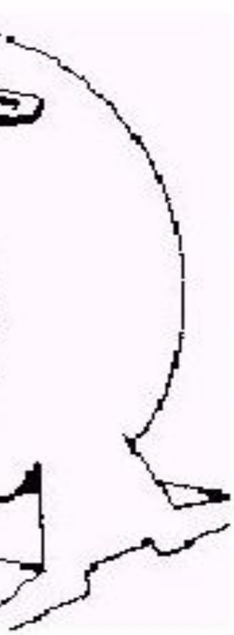
It is the outer cover of machine.

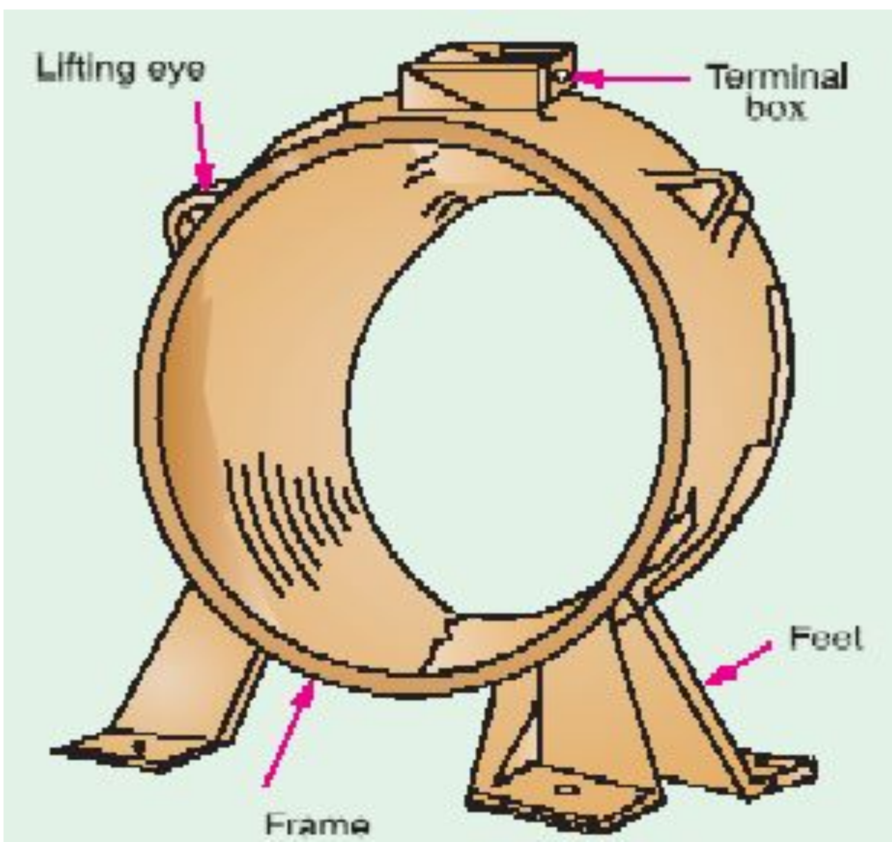
oose of outermost cover of the dc machine. So that
: protected from moisture, dust etc.
al support to the poles.
of low reluctance for the magnetic flux produced by

al:
up of some magnetic material to provide a low
nagnetic flux.
s cast iron is used for the construction because it is

s rolled steel, cast steel etc are used.

nstructional Details





nstructional Details

ield system is the stationary part of the machine uces the magnetic flux. It includes pole core, pole indings.

achine has even number of poles. Pole cores are

ally carries field winding which is necessary to

rial:
f steel laminations.

nstructional Details

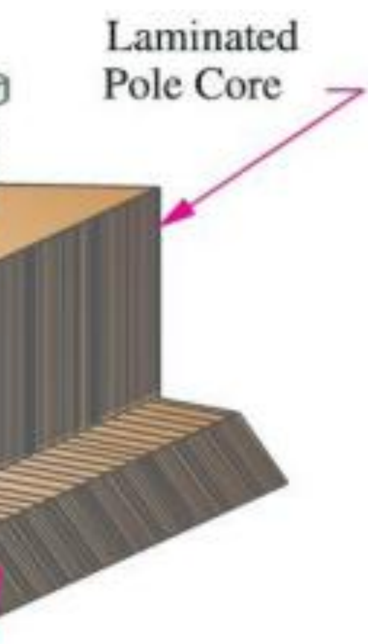
e core has a pole shoe of curved surface.

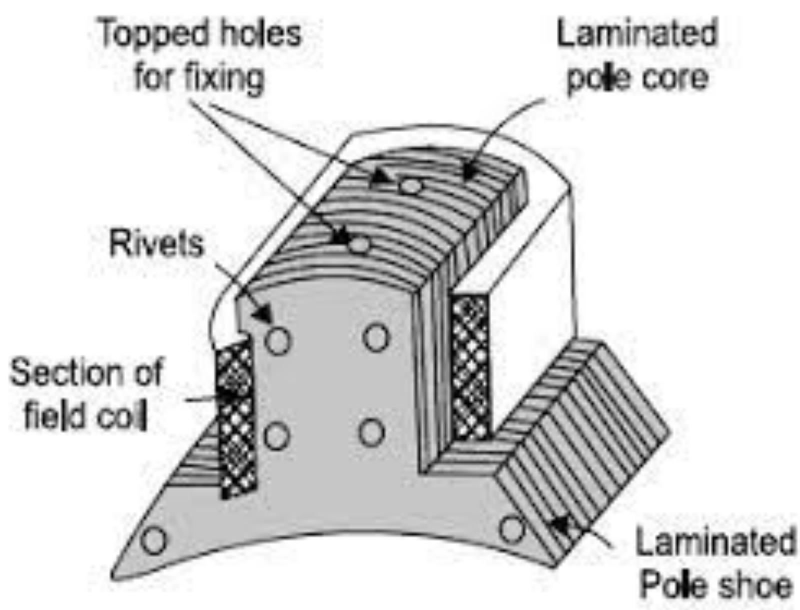
rt the field windings
ux in air gap uniformly
ss-sectional area of the magnetic circuit and thus
ance of the flux path.

ial:
f steel laminations.

nstructional Details

Shoe:





nstructional Details

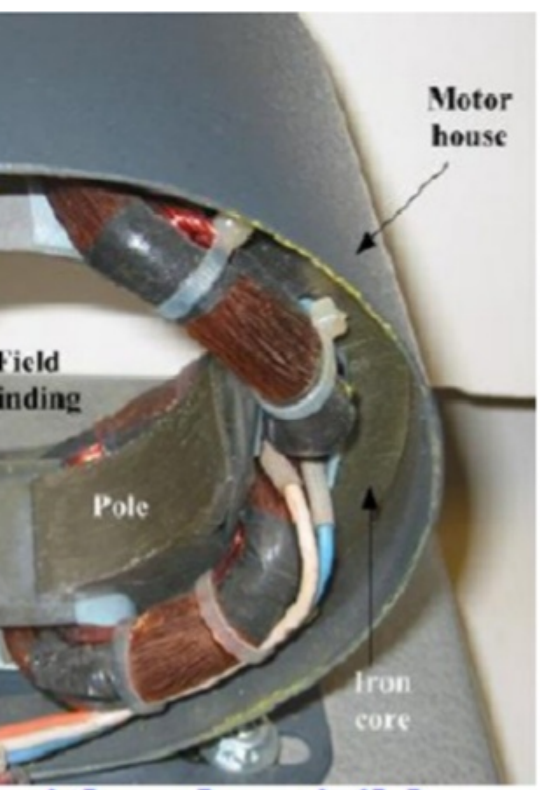
windings are wound on the pole core to produce are also called pole coils or exciting coils. These coils poles in such a direction so as to form alternate north es.

tromagnets are used for producing necessary flux. field windings carry current, pole cores get nd produce flux.

al:

e construction of field windings.

nstructional Details



nstructional Details

ature is the rotating part of the machine. It mainly
) Armature core b) Armature windings

s cylindrical in shape and mounted on the shaft. It
s outer periphery.

e windings on the slots provided on its periphery and
otate in the magnetic field.
of low reluctance for the magnetic flux produced by

al:

low reluctance path for the flux, it is made of cast
is made of laminations to reduce eddy current loss.

nstructional Details

: These are placed in the slots provided on the
ure core. Armature windings are of two types: Lap
winding. Their selection will depend on the current
e.

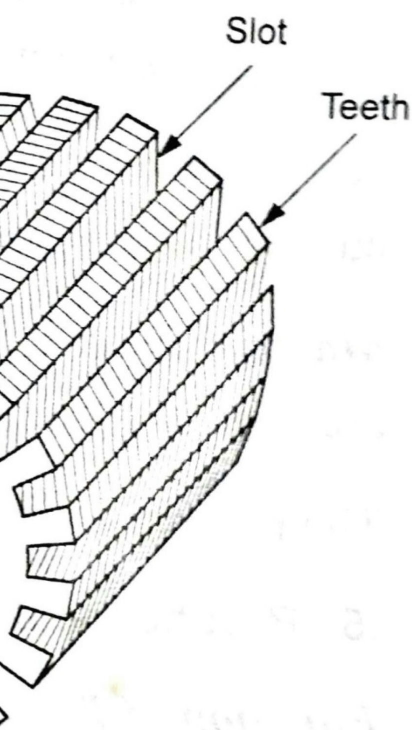
tor, generation of EMF takes place in the armature

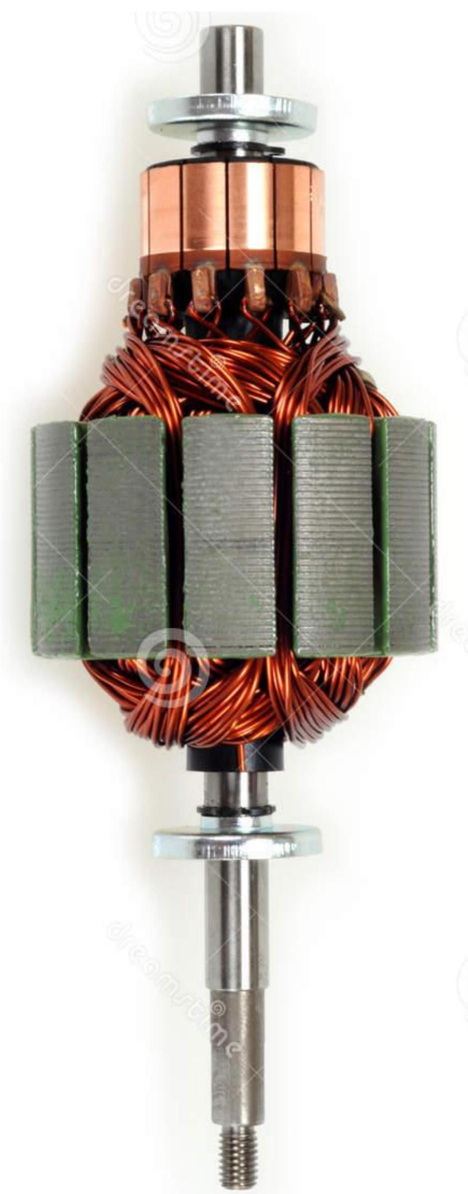
it carries the supply current.

al:

re made of Copper.

nstructional Details





nstructional Details

he commutator is of cylindrical in shape and number of segments. Number of segments is umber of armature coils.

of the commutator is to convert alternating in the armature into unidirectional current.

al:

de of wedge-shaped hard drawn forged copper. ents are insulated from each other with thin .

nstructional Details

nstructional Details

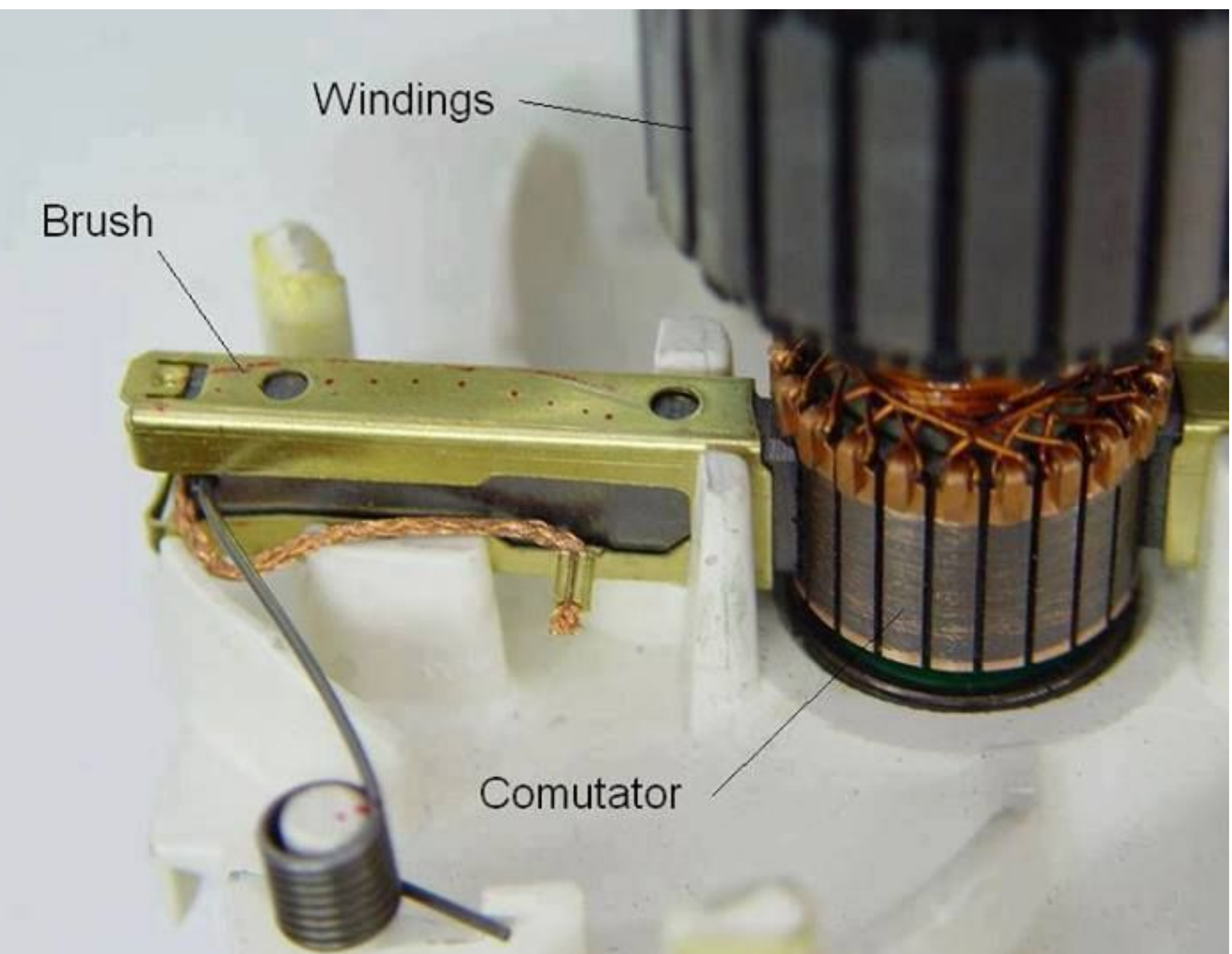
rushes are placed in brush holders. They rest on tor segments. The commutator and brush s the connecting link between the armature xternal circuit.

ent from the commutator of the machine

al:

of Carbon.

nstructional Details



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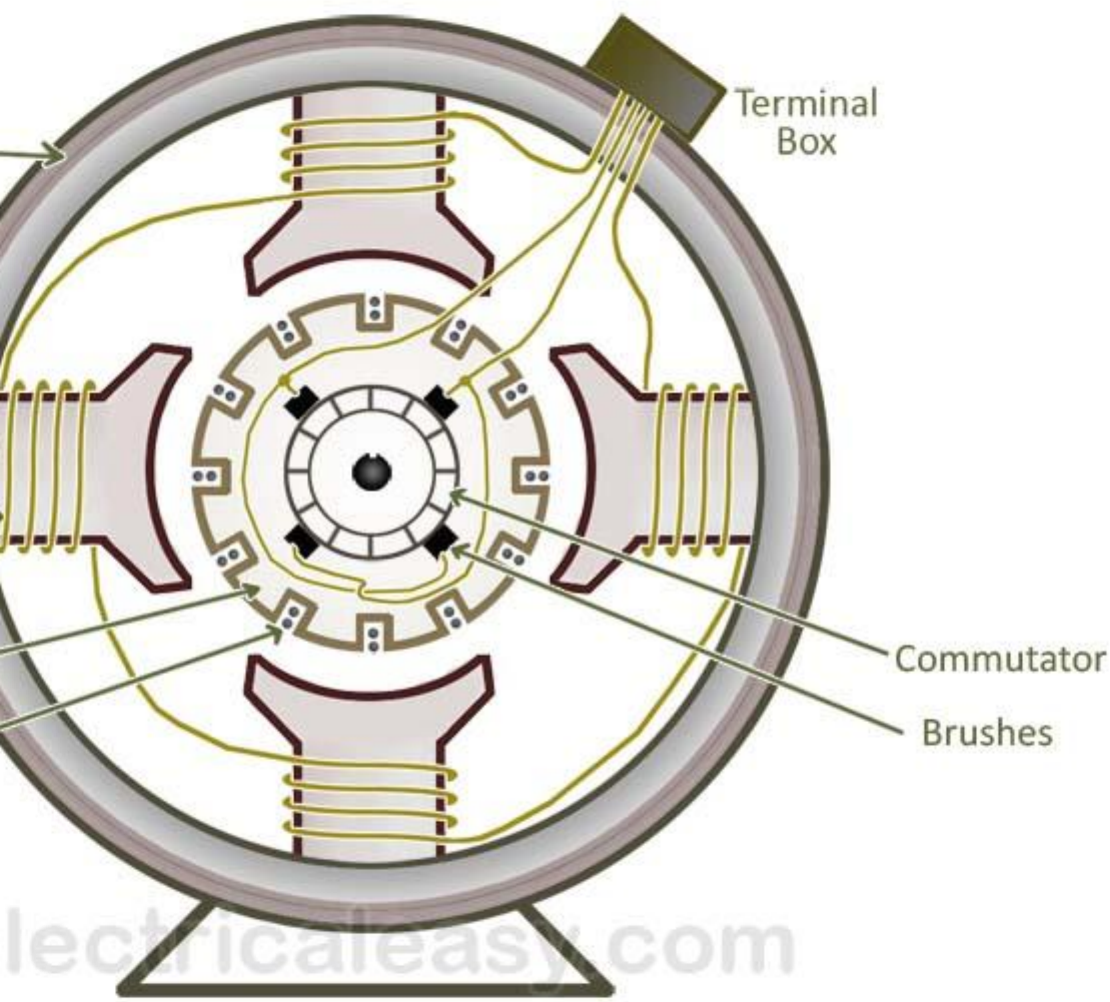

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